

Day 3 Meeting Summaries

Loredo, 1pm EST Live Q&A

Quick recap

Tom Laredo conducted a Q&A session following his lectures on Bayesian data analysis, where he discussed key takeaways including the distinction between probability and frequency, and the importance of marginalization in Bayesian inference beyond just Bayes' theorem. He addressed Jessica's question about where prior prevalence data comes from in disease classification problems, explaining how hierarchical Bayesian modeling can handle uncertainty in prevalence estimates. Breanna asked about incorporating machine learning probabilities into Bayesian analysis, which led to a discussion about interim priors and the need to report likelihoods rather than just posterior probabilities for scientific data sharing. Tom offered to provide individual support to participants interested in Bayesian applications, and George inquired about applying Bayesian methods to the Drake equation, which Tom said would be an interesting application of hierarchical modeling to quantitatively handle the many unknown parameters.

Next steps

Tom

- Post references on hierarchical Bayesian modeling and cosmic demographics (including the 2004 tutorial paper and the more rigorous paper with mathematical appendix) in Slack for attendees.
- Dig into and report back on Slack about any existing Bayesian treatments or papers related to the Drake equation, as discussed in response to George's question.
- Continue to monitor and answer questions on Slack related to Bayesian data analysis.
- Offer and be available for one-on-one discussions (via email/Zoom) with attendees interested in Bayesian methods for their specific problems.

Summary

Bayesian Data Analysis Overview

Tom discussed Bayesian data analysis and highlighted key takeaways from the lectures. He emphasized the distinction between probability and frequency as a fundamental concept in Bayesian reasoning. Tom announced that the lecture videos would remain available on

the Cornell Video on Demand service through July for those who haven't had time to review them.

Bayesian Probability and Theory

Tom discussed the historical relationship between probability and frequency, explaining how early mathematicians like Laplace and Bernoulli developed Bayesian perspectives on probability. He described how Thomas Bayes provided the first application of Bayes' theorem to show the connection between observed frequencies and probabilities of single trials. Tom also explained modern philosophy of science's shift from Popper's falsifiability theory to confirmation theory, which uses Bayesian notions of probability, and recommended the book "10 Great Ideas About Chance" by Diaconis and Skirms for further reading. Finally, Tom emphasized that Bayesian data analysis involves more than just Bayes' theorem, with subsequent steps primarily using the law of total probability or marginalization.

Bayesian Inference in Machine Learning

Tom discussed Bayesian inference, highlighting its importance in marginalization and its relevance to machine learning, particularly deep learning. He explained how Bayesian inference allows for marginalization over parameters, which is not possible in frequentist statistics. When addressing a question about the source of prior prevalence in a disease diagnosis example, Tom explained that it could come from medical studies or trials, and in cases with uncertain data, hierarchical Bayesian modeling or graphical modeling might be used to estimate and account for the uncertainty in prevalence.

Hierarchical Bayesian Data Analysis

Tom explained hierarchical Bayesian data analysis using medical trials as an example and applied the concept to cosmic demographics, such as classifying stars and galaxies in astronomy. He discussed how this method allows for simultaneous inference of prevalence and individual classifications when data is pooled. Tom mentioned his 2004 tutorial paper on Bayesian cosmic demographics, which gained attention in recent years due to its relevance in gravitational wave data analysis. The conversation ended with technical difficulties regarding Breanna's audio.

Machine Learning and Bayesian Analysis

Tom and Breanna discussed the challenges of incorporating machine learning probabilities into Bayesian analysis for astronomical source classification. Tom explained the concept of interim priors and the importance of reporting likelihoods rather than just posterior probabilities, particularly for scientific data that may be reused in ongoing discovery

processes. He offered to discuss this topic further offline and invited Breanna to reach out via email if she wants to explore Bayesian methods for her classification work.

Bayesian Methods and Drake Equation

Tom discussed the potential application of Bayesian methods to the Drake equation, noting that while some parameters like planet prevalence are now known, others remain uncertain. He mentioned a German physicist Volker Dose who introduced Bayesian methods into plasma physics and recalled discussing the Drake equation from a Bayesian perspective during Dose's retirement talk. Tom agreed to research and share information about Bayesian approaches to the Drake equation on Slack and mentioned that Murali Horan would be giving a talk on Markov Chain Monte Carlo methods in the future.

Haran, 3:20pm EST Live Q&A

Quick recap

This was an interactive Q&A session where Murali answered questions about MCMC (Markov Chain Monte Carlo) methods and Bayesian inference tools. Tom asked about the frequency of using standard MCMC packages versus writing bespoke code, to which Murali responded that he typically starts with packages like Stan, Nimble, or PyMC for applied work, but moves to custom implementations only when dealing with high-dimensional matrix operations or intractable likelihood problems. Hyungsuk inquired about diagnosing convergence in MCMC chains, and Murali explained key diagnostics including checking MCMC standard errors, examining trace plots to assess starting value impact, and ensuring adequate effective sample sizes (typically 4,000 per parameter). The discussion concluded with Hyungsuk asking about the future role of MCMC in the era of simulation-based inference (SBI) and nested sampling, where Murali noted that while SBI has opened new possibilities for previously intractable problems, MCMC still offers theoretical guarantees and reliability that make it preferable when both approaches are viable.

Next steps

Next steps were not generated due to insufficient transcript.

Summary

Tutorial Support Meeting Discussion

Murali hosted a meeting and waited for questions about a tutorial, offering to address any issues participants encountered. Several participants, including Hyungsuk and Tak, joined the meeting, with some experiencing technical difficulties with audio and video. Tom asked a question at the end of the transcript, though the specific content of the question was not captured.

Environmental Science Modeling Tools Discussion

Murali explained that for applied work in environmental sciences, he typically starts with using Stan, Nimble, or PyMC packages, as they are easier to use and extend. He noted that he would only write custom MCMC code when dealing with high-dimensional matrix operations or intractable likelihood problems, which are still challenging to handle in the standard packages. Murali emphasized the benefits of using high-level packages, particularly in terms of efficiency and reducing debugging time.

Intractable Likelihood Problems in Statistics

Murali discussed the evolution of addressing intractable likelihood problems in statistics, noting that while bespoke solutions were once common, simulation-based inference packages like SBI have become more prevalent, especially in astro. He emphasized that understanding the theoretical foundations of MCMC and Monte Carlo methods is now more important than ever, as there is more available code that generates Markov chains. Murali stressed that statisticians need to understand the underlying theory to ensure proper scientific methodology when using these algorithms.

MCMC Sampling Diagnostic Methods

Murali explained diagnostic methods for MCMC sampling, focusing on two key areas: evaluating the impact of starting values on posterior approximations and assessing the variability of those approximations using MCMC standard errors. He emphasized the importance of checking trace plots to ensure the Markov chain reaches a stable region and recommended looking for an effective sample size (ESS) of at least 4,000 for each parameter as a heuristic guideline. Murali warned against arbitrarily discarding initial samples as it would increase the standard error of approximations, noting the inherent bias-variance trade-off in statistical methods.

ESS and Convergence Diagnostic Tools

Murali confirmed that ESS can be different for different parameters, using 4,000 as a heuristic for each parameter. Tom asked about using trace plots of log likelihood or log posterior as diagnostic tools for convergence issues, and Murali suggested it could be

useful for identifying when parameters are "hanging out on a ridge" but noted there's no established mathematical framework for using these plots to diagnose convergence.

MCMC and SBI Discussion

Murali and Hyungsuk discussed the role of Markov Chain Monte Carlo (MCMC) in the era of various Bayesian inference tools, including simulation-based inference (SBI) and nested sampling. Murali explained that while SBI offers new capabilities for solving previously unsolvable problems, MCMC still provides theoretical guarantees and can be more reliable for certain types of problems. He indicated that he plans to continue using both MCMC and SBI in his work going forward.

Speagle, 5:30pm EST Live Q&A

Quick recap

Joshua conducted a Q&A session about nested sampling, where he explained the key differences between MCMC and nested sampling methods. Marios asked how nested sampling can provide finite-time guarantees unlike MCMC, and Joshua responded that nested sampling is designed to shrink the parameter space progressively until it reaches a point where additional sampling doesn't provide meaningful information, whereas MCMC requires infinite time for convergence. Joshua noted that while many participants were new to Bayesian methods and nested sampling, the approach is particularly useful for problems with multiple solutions, complex uncertainty distributions, and model comparison scenarios where computing marginal likelihoods is important. He provided a practical example using gravitational wave signals with circular uncertainties, explaining how nested sampling can be more efficient than MCMC in handling such degeneracies by proposing samples across the entire parameter space rather than starting from a single position.

Next steps

Next steps were not generated due to insufficient transcript.

Summary

Astrostatistics Workshop Introduction Session

Joshua introduced himself as an assistant professor at the University of Toronto specializing in astrostatistics and welcomed participants to ask questions about nested sampling for the next 20 minutes. He indicated he would be available to answer questions about both traditional statistical methods and machine learning approaches covered in the

workshop. Joshua requested participants to either raise their hands or post in the chat with questions, and offered to navigate to specific slides if provided with the slide numbers.

MCMC vs Nested Sampling Methods

Marios asked Joshua for clarification on the differences between MCMC and nested sampling methods, particularly regarding their finite sample guarantees. Joshua explained that MCMC operates as an infinite-time algorithm relying on the convergence of a Markov chain, while nested sampling is designed to provide finite-time guarantees through its shrinking process that eventually reaches a point where further sampling doesn't provide additional information. Joshua also mentioned that there are finite-time versions of MCMC methods that use pairs of chains to meet and join together, providing some hybrid approaches between the two methods.

Nested Sampling Methods in Astrostatistics

Joshua led a discussion about nested sampling methods in astrostatistics, explaining its advantages over standard MCMC techniques, particularly in handling multimodal distributions and complex uncertainty patterns like rings or banana shapes. He provided a practical example of how nested sampling can efficiently handle circular uncertainties by proposing points from the entire space rather than starting from a single position and moving slowly. Joshua encouraged participants to reach out with further questions on Slack and mentioned upcoming examples and code implementation opportunities.